LAB REPORT

CSE312: Database Management System Lab

Submitted To
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Submitted By
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Experiment No: 03	Experiment Name: Performing DQL Operations on the national Database Using SQL Queries

1. Introduction

This experiment was conducted using the provided national.sql database to practice SQL retrieval operations in a relational database system. The tasks focused on searching data with subqueries, comparing values with average population, and grouping records to produce summarized results. This lab helped develop practical understanding of SQL query writing, relational thinking, and result analysis using phpMyAdmin.

2. Objectives

The main objectives of this experiment were:

- To retrieve data from related tables using SQL queries.
- To use **subqueries** for solving relational data problems.
- To apply **aggregate functions** such as AVG() and COUNT().
- To use **GROUP BY** for summarized region-based reports.
- To improve practical skills in writing and testing SQL queries in a database environment.

3. Assigned Tasks

The following tasks were assigned in this lab:

1. Display all nations whose region_id is the same as Germany's region.
2. Show the name of the nation that belongs to the region named **Asia** using a subquery.
3. Show all data of those countries whose population is higher than the average population.
4. Find the total number of countries in every region in this database.

4. Software and Tools Used

- **Web Server Stack:** XAMPP Control Panel v3.3.0
- **Database Management System:** MariaDB / MySQL
- **Database Interface Tool:** phpMyAdmin
- **Database File:** national.sql
- **Text Editor:** Notepad (232-15-818.txt)
- **Operating System:** Windows

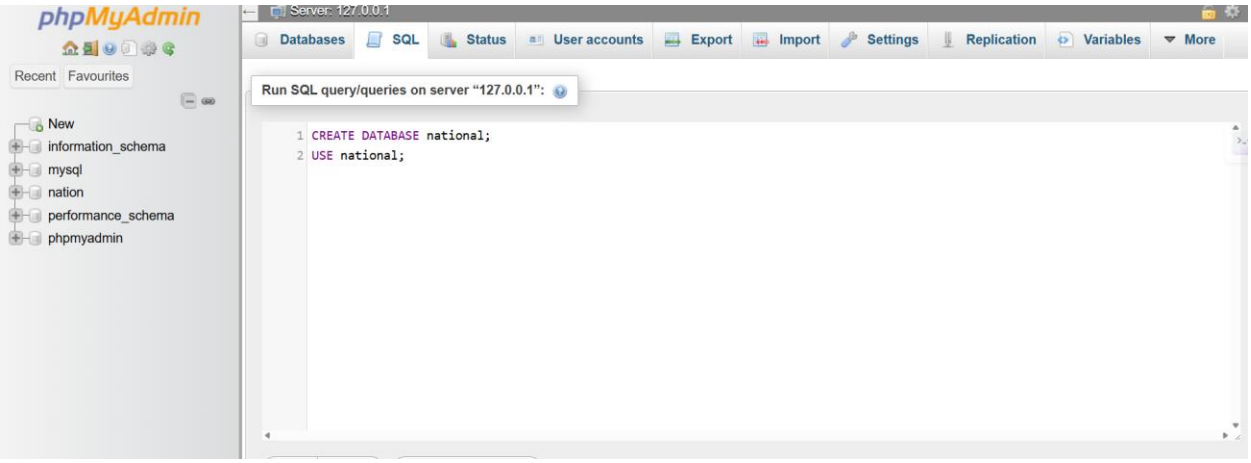
5. Methodology

The experiment was completed by following these steps:

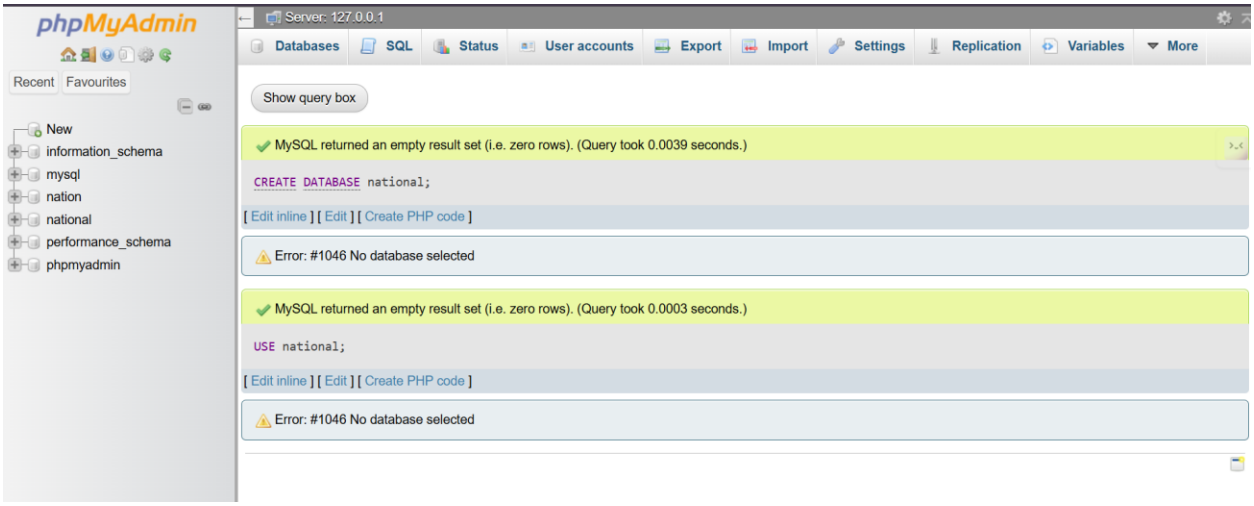
1. XAMPP was started and phpMyAdmin was opened.
2. The national database was created and the provided national.sql file was imported.
3. The nation and region tables were reviewed to understand their structure and relationships.
4. The assigned questions were analyzed carefully before writing SQL statements.
5. Required SQL queries were written and executed one by one.
6. The output of each query was checked and verified.
7. Screenshots of queries and outputs were collected for report preparation.
8. The final report was organized in a structured academic format.

6. Experiment Solutions:

Step-01: Create national Database



Step-02: Success of Compilation (national)

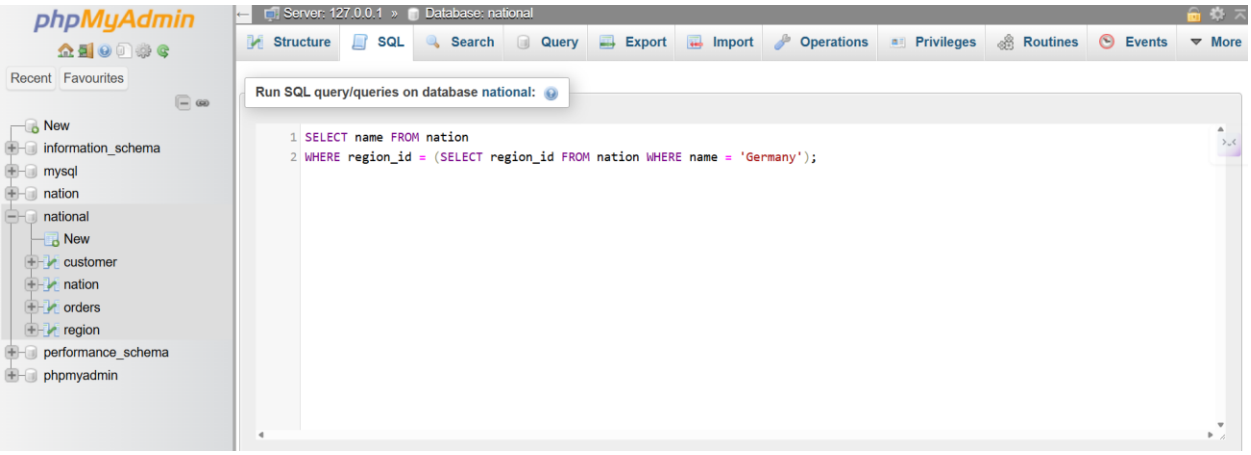


Step-03: Import national.sql file

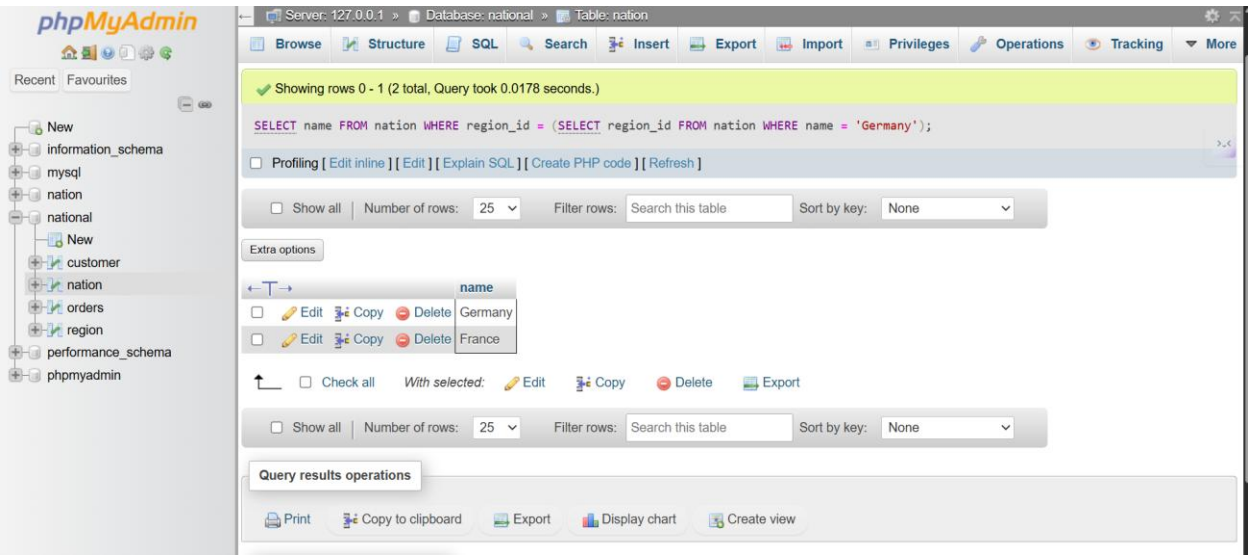
DQL Operations and SQL Queries

Task 6.1: Display all nations whose region id is the same as Germany's region.

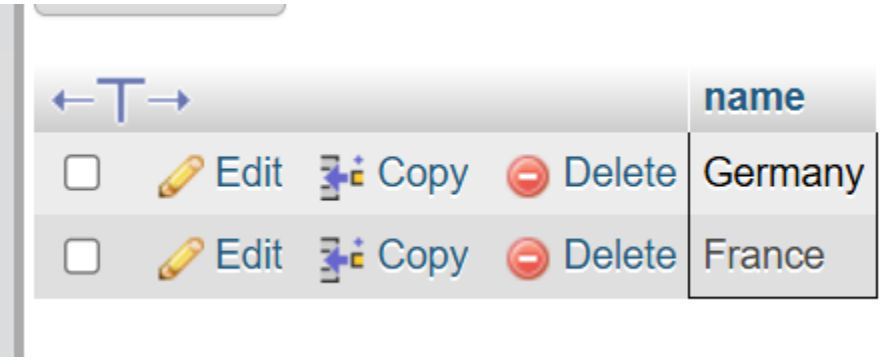
Step-01: SQL Query



Step-02: Success of Compilation

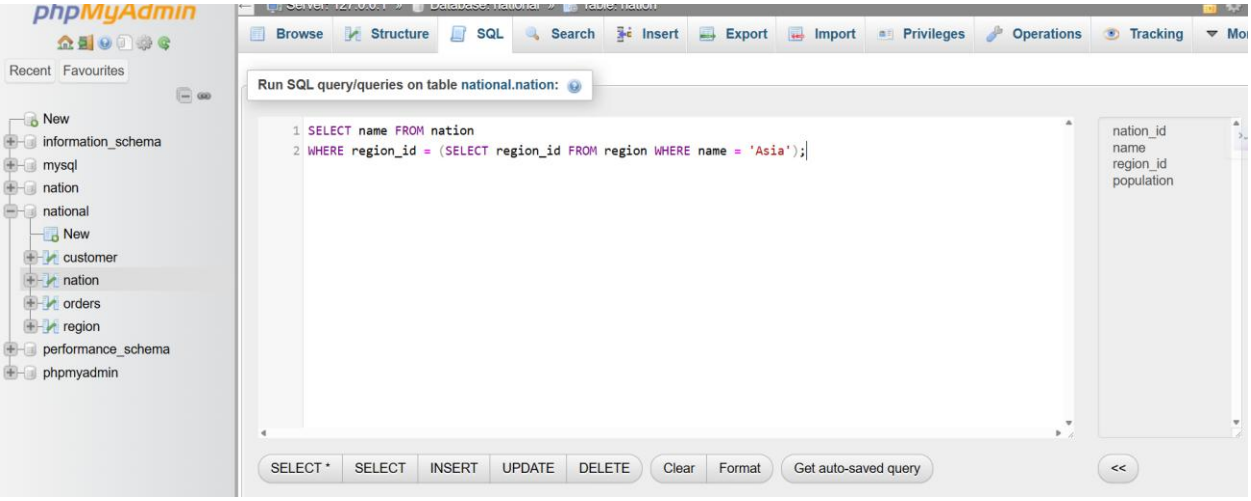


Step-03: Output

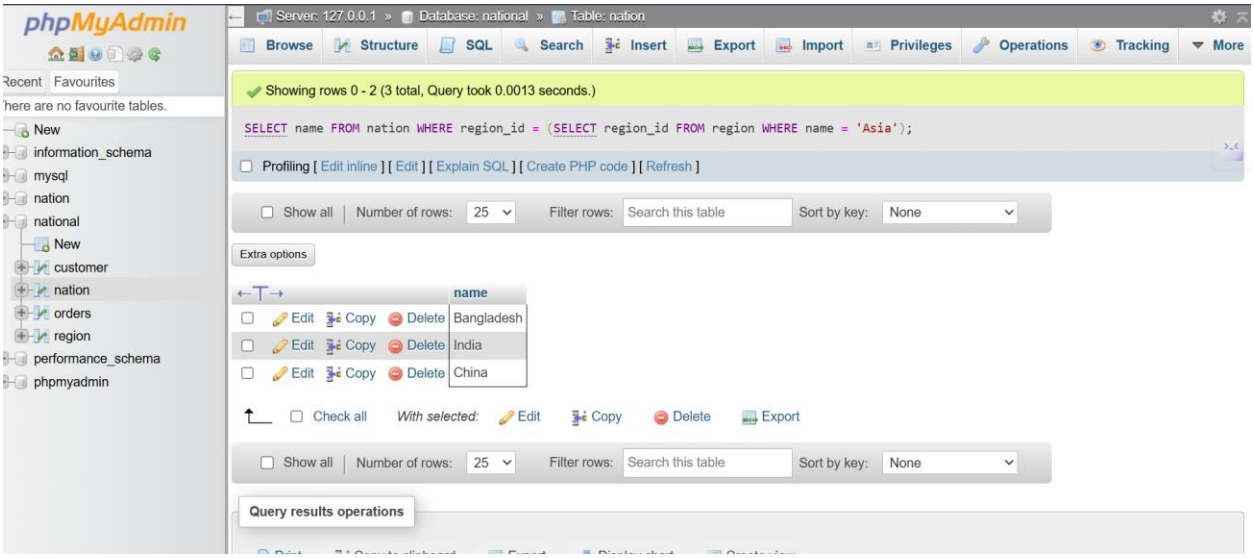


<u>Task 6.2:</u> Show the name of the nation that belongs to the region named 'Asia' using a subquery.

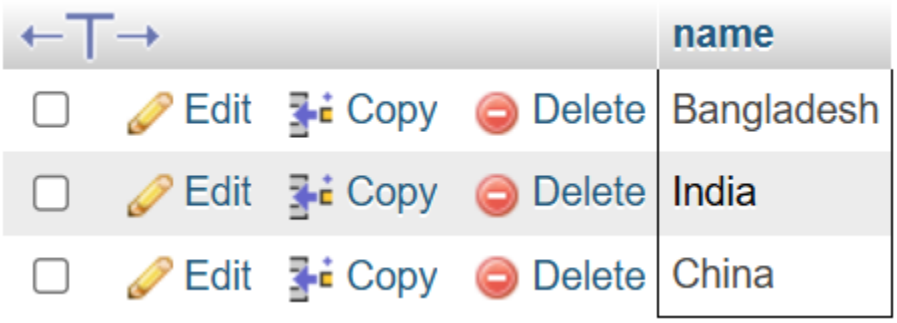
Step-01: SQL Query



Step-02: Success of Compilation

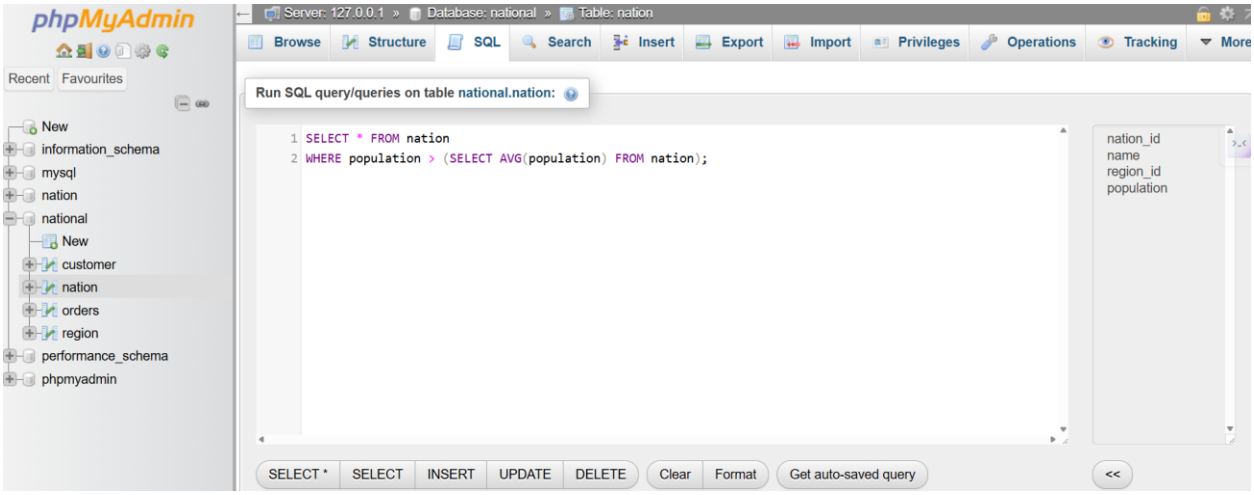


Step-03: Output

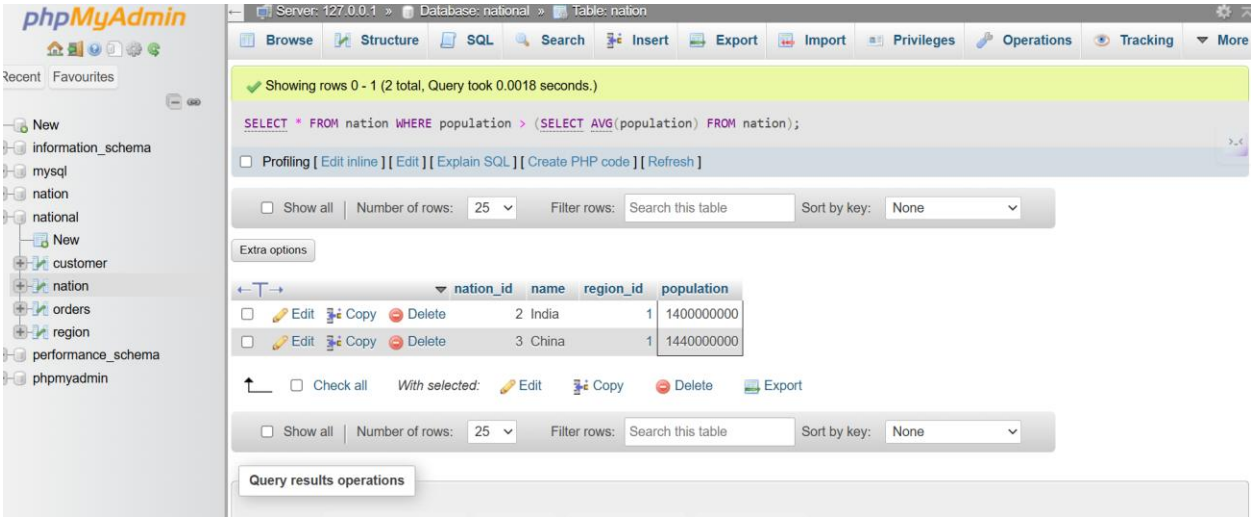


<u>Task 6.3:</u> Show all the data of those countries which has higher population than the average population.

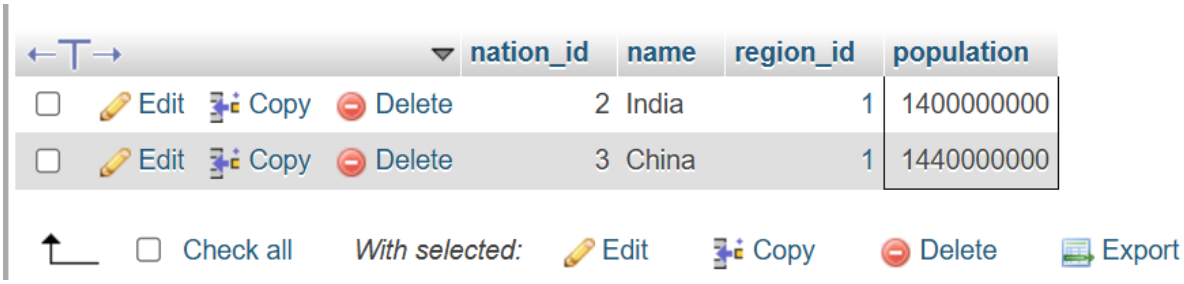
Step-01: SQL Query



Step-02: Success of Compilation



Step-03: Output

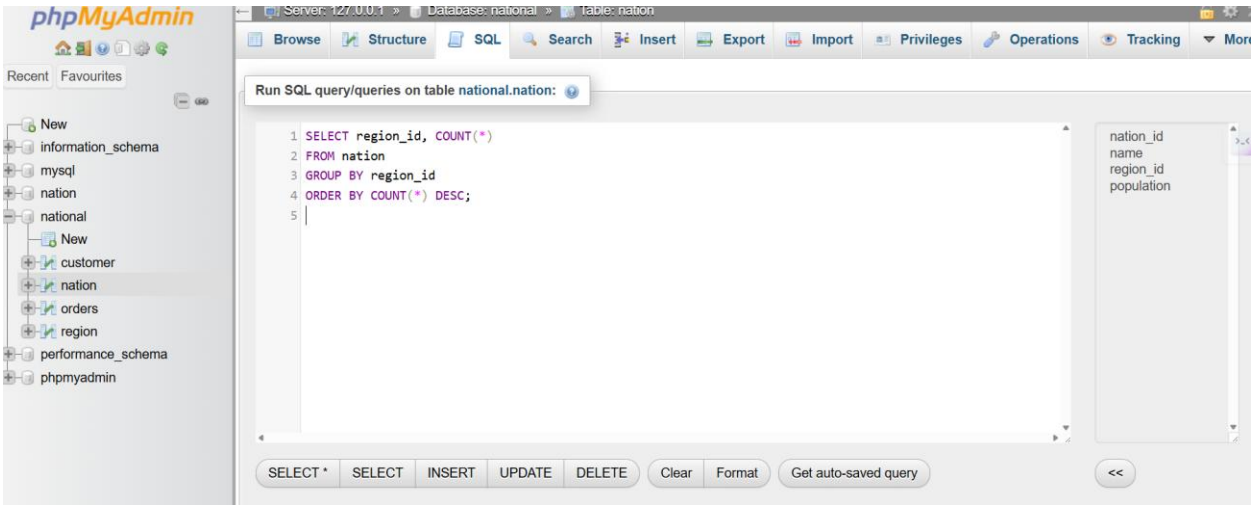


	nation_id	name	region_id	population
Edit Copy Delete	2	India	1	1400000000
Edit Copy Delete	3	China	1	1440000000

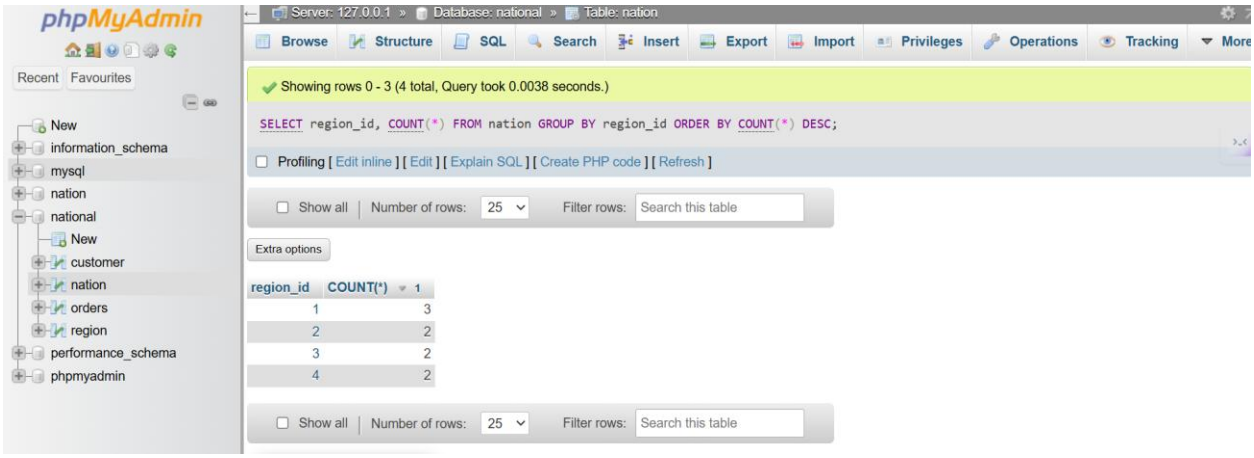
Check all *With selected:* Edit Copy Delete Export

<u>Task 6.4:</u> Find the total number of countries in every region in this database.

Step-01: SQL Query



Step-02: Success of Compilation



Step-03: Output

Extra options

region_id	COUNT(*) 1
1	3
2	2
3	2
4	2

Show all | Number of rows: 25 Filter rows: Search this table

7. Observation

From this experiment, the following observations were made:

- Subqueries are very useful when one query depends on the output of another query.
- The **AVG()** function helps compare a row value with the average of the full table.
- The **COUNT()** function with **GROUP BY** is effective for summary reporting.
- The relation between region and nation tables makes SQL data retrieval easier and more meaningful.
- SQL can solve both detailed record-based queries and summarized analytical queries in a simple way.

8. Conclusion

In conclusion, this experiment improved practical knowledge of SQL data retrieval techniques using subqueries, aggregate functions, and grouping. By solving real queries on the national database, the lab strengthened understanding of relational database concepts and query logic. This experiment was helpful for learning how to extract meaningful information from related tables in an organized and efficient way.

9. Appendix

Appendix A: Course Outcome (CO) Attainment

SN	COs	Attainment	Justification
01	CO1	Yes	The lab required writing and executing SQL queries to retrieve and analyze database records.
02	CO2	Yes	The experiment used subqueries, aggregation, and grouping on relational tables.
03	CO3	No	Advanced database administration and transaction management were not covered in this experiment.

Appendix B: Addressing Complex Engineering Problems (EP)

SN	Engineering Problem (EP)	Attainment	Related CO	Justification
01	EP1: Depth of Knowledge	Yes	CO1, CO2	The experiment required understanding of SQL syntax, relational structure, and analytical queries.
02	EP2: Conflicting Requirements	No	CO2	The tasks were direct and did not involve conflicting requirements.
03	EP4: Familiarity of Issues	Yes	CO1	The experiment covered standard academic DBMS lab operations.
04	EP7: Interdependence	Yes	CO2	The tasks depended on the relationship between `region` and `nation` tables.